



Science for a
moving society

LDCI2027: Light-Duty CI-ICE 2027

IGF 366 EN/2 | FVV No. 1507

PA4 – WebCo

Prof. Dr. Cecilio Angulo (UPC)

Markus Frey (IFS)

Dr.-Ing. Ferhat Inci (FZA)

Introduction

COMPLIANCE RULES



Compliance



We would like to draw your attention to national and European competition law that prohibits discussions on competitive topics, including prices or discounts, or the exchange of other sensitive company data during association meetings. The prohibition also extends to the coordination of industry practices and the making of decisions and agreements to such effect. Any such behaviour is susceptible to severe penalties that will be imposed on the association and its member companies. Thus, adherence to these basic rules is essential.

→ FVV CODE OF CONDUCT

Introduction

MEETING MINUTES

Meeting minutes ([Link](#))

File: Prot_1507_PA3-WebCo_2025-04-04.pdf



MINUTES



FVV project 1507 “LDCI2027 Light-duty CI-ICE 2027” (PA3)

Date 04.04.2025
Time 09:30 – 14:30 hrs (CEST)
Venue Webmeeting
Minutes Frank Bunar
Participant see attached list of participants

Date of next meeting | 10.10.2025

TOP 1 Compliance

- Participants are advised of the FVV Code of Conduct and the compliance rules applicable to an FVV event. The Code is filed as a document in [THEMIS](#).

TOP 2 Review meeting minutes of last meeting

- No comments or changes requested by the working group

TOP 3 Presentation of project contents by the research institutes // TU Berlin by Ferhat Inci

LDCI2027: Light-Duty CI-ICE 2027
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PA3 – WebCo
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Presentation:

File: PR_1507_PA3-WebCo_FZA_2025-04-04.pdf

- Review
 - Key results of PA3
 - Goals until PA4
- Overview/Schedule
 - Project Overview
 - Review/Discussion Research Aspects
 - Current status of project
- Working Packages
 - University of Stuttgart
 - Technische Universität Berlin
 - Universitat Politècnica de Catalunya
- Next Steps
 - Goals until the next PA
 - Next PA Meeting
- Feedback
 - Possible Future Research Aspects
 - Comments/Discussion/Questions

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Review

KEY RESULTS OF PA3

- Results of reference system measurements
 - Engine map, warm-up
 - DoE for engine model set-up was measured
- Set-up of pre-turbo system was completed
 - Adaption of bypass adapter needed
- Data model validation
- LSTM-MBPC control architecture completed
- RL Recurrent TD3 control architecture is defined

Review

GOALS UNTIL PA4



Technische Universität Berlin



- Completion of reference WLTC measurements
- Conversion and commissioning of Pre-Turbo-System ✓
- Securing test environment for AI testing ✗

→ *Discussion adaption of project targets towards technology verification*

University of Stuttgart

- XXX

Universitat Politècnica de Catalunya

- Extending LSTM-MBPC control architecture to new situations
- Fitting RL Recurrent TD3 control architecture

→ *Extend MBPC to more complete controllers / Go in depth with RL*

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Overview/Schedule

PROJECT OVERVIEW

Test Bench

- Engine Measurement for calibrating simulation models
- Testing final AI – control strategy

Physical Simulation Model

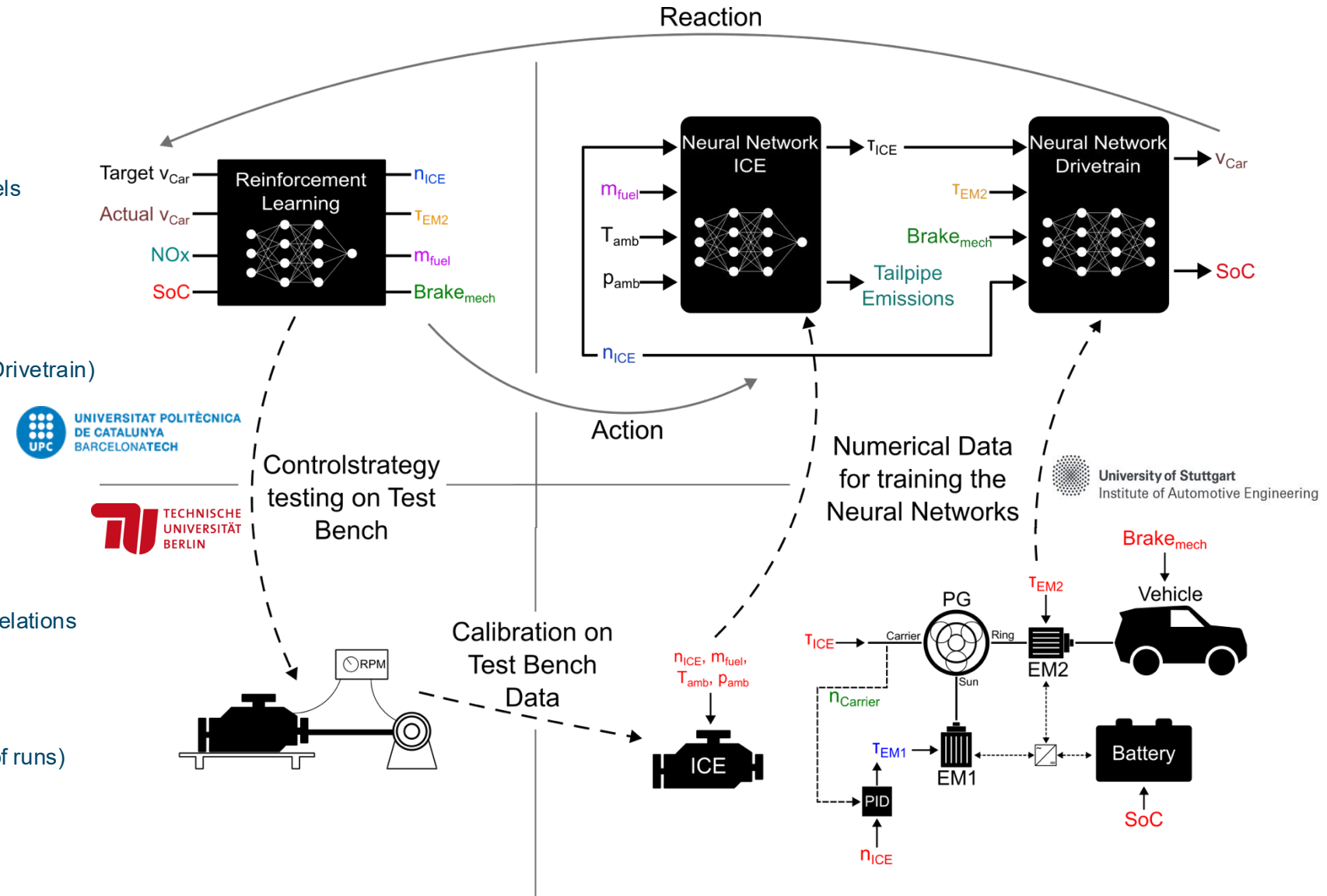
- Precise representation of a physical system (ICE + Drivetrain)

AI – Model

- Ultra fast running environment (faster than real time)
- High accurate system representation
- Representation of high non-linear and complex correlations

Reinforcement Learning Algorithm

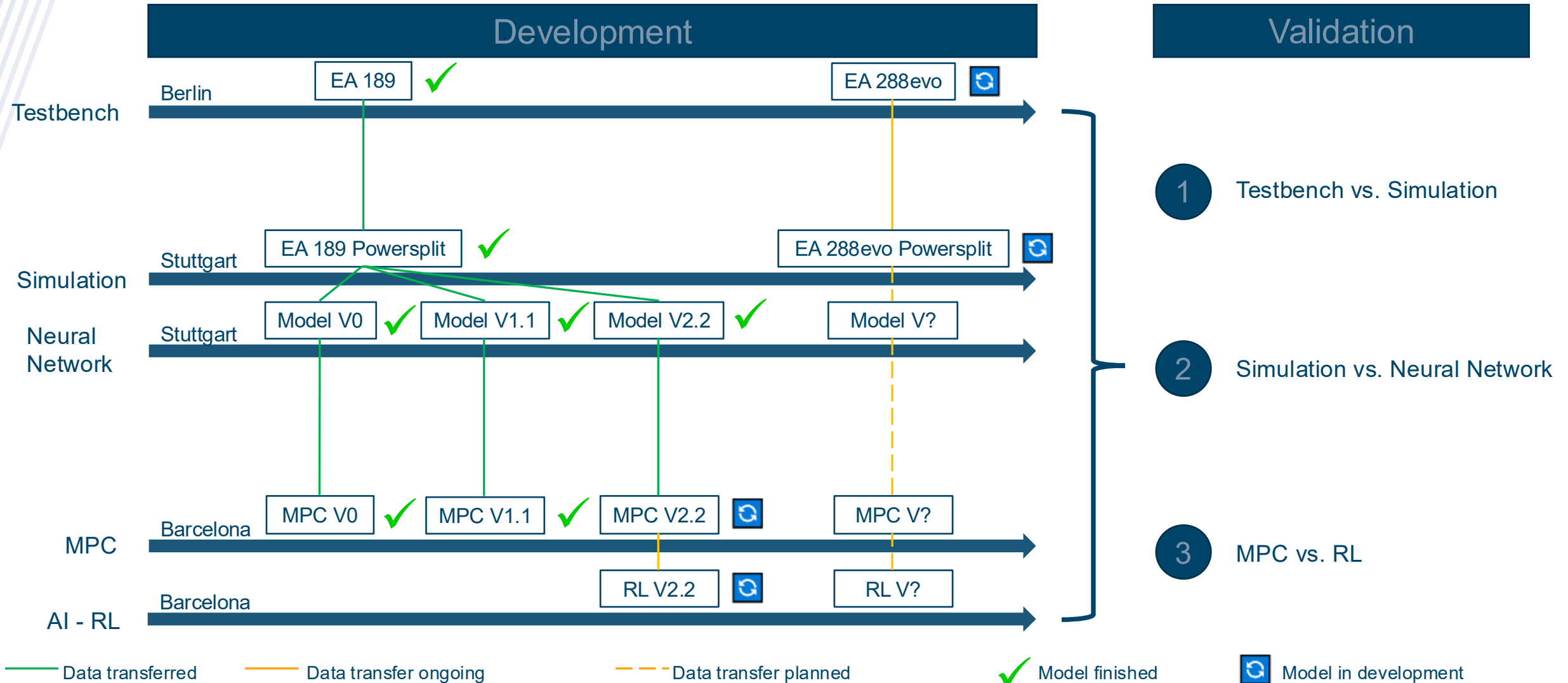
- Optimization of a problem by try and error (millions of runs)
- Interaction with an environment, that describes the underlying system (AI – Model)



Overview/Schedule

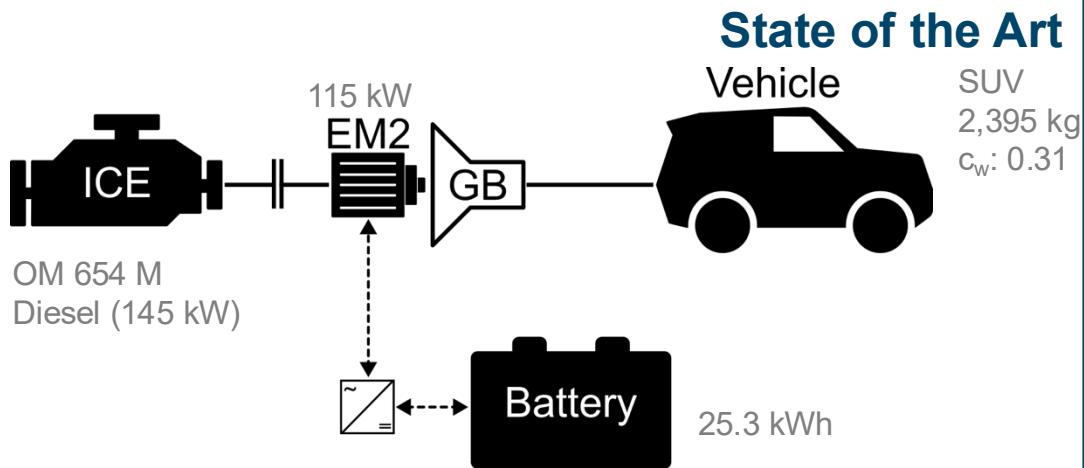


PROJECT OVERVIEW

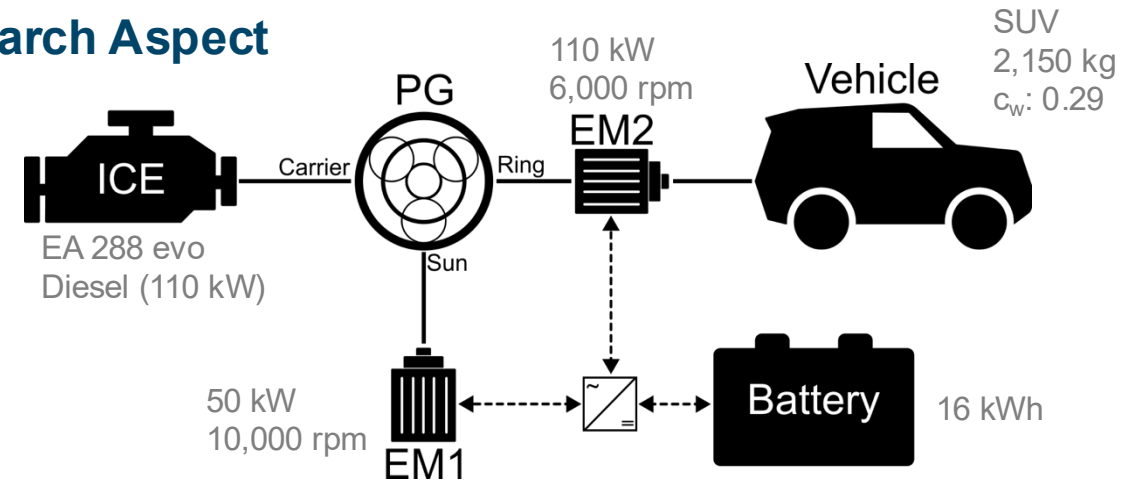


Overview/Schedule

REVIEW/DISCUSSION RESEARCH ASPECTS - POWERTRAIN



Research Aspect



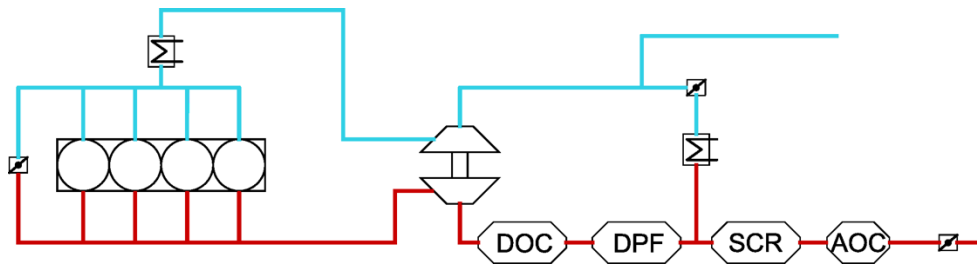
- Electrification is a major topic, and **market demand is currently shifting towards hybrids**
- Power-split hybrids on the market gasoline-based and only in the passenger car segment available (Toyota Prius)
- Diesel PHEV in the SUV/light-duty segment are mainly parallel systems (Mercedes GLC 300de)

- Combination of diesel engine with powersplit hybrid drivetrain offers high flexibility
- Dual e-machines enable torque split, planetary gear offers continuously variable transmission and a fast response
- Diesel powersplit hybridization is a **completely new concept**. With the current rise in hybrid demand, this research aspect gains **increasing relevance and strong potential** for future applications

Overview/Schedule

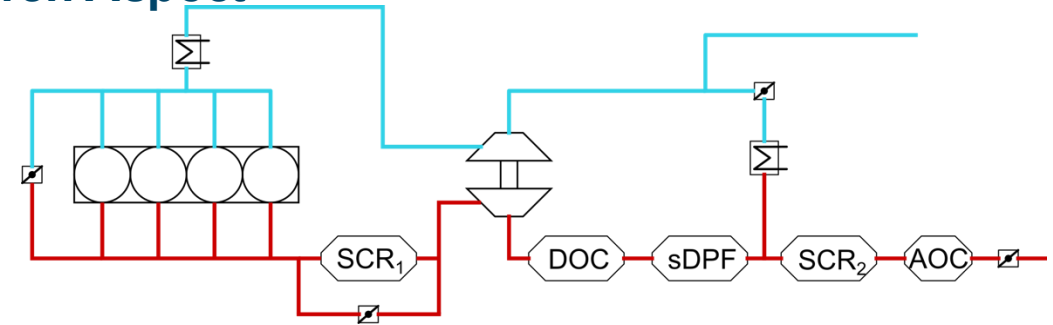
REVIEW/DISCUSSION RESEARCH ASPECTS – AFTERTREATMENT SETUP

State of the Art Research Aspect



- Typical aftertreatment system of EA288evo consist of DOC, DPF, SCR and AOC
- All placed downstream of the turbine
- Fine for Euro 6 & 7 (falling unexpected short)
- Current EA 288evo aftertreatment setup with DOC–DPF–SCR–AOC meets Euro 6/7 but **suffers from long heat-up times and limited cold start efficiency.**

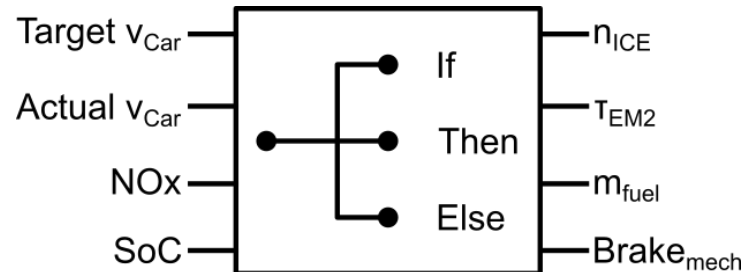
DOC: Diesel Oxidation Catalyst; DPF: Diesel Particulate Filter; SCR: Selective Catalytic Reduction; AOC: Ammonia Oxidation Catalyst



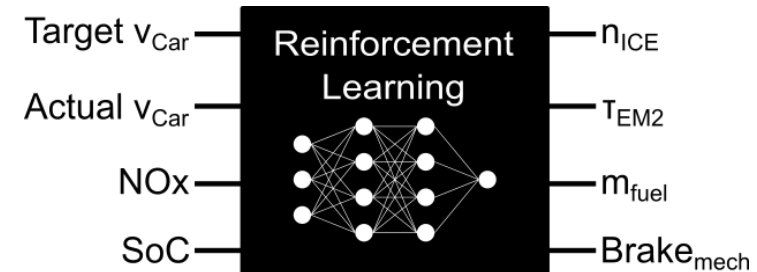
- **Post Euro 7 scenario:** stricter emission limits especially during cold start
- FVV Project 1305 investigated several pre-turbine setups to decrease time to heat up und increase conversion efficiency
- Adopted a setup with SCR pre turbine to improve NOx conversion already during cold start
- Novel pre-turbine SCR layout enables earlier NOx conversion and **compliance with post Euro 7 scenarios cold-start limits**, but increases back pressure and **reduces dynamic performance**

REVIEW/DISCUSSION RESEARCH ASPECTS - CONTROL STRATEGY

State of the Art | Research Aspect



- Conventional rule-based control strategies
- Predefined operating maps and heuristics
- Limited adaptability to varying driving conditions
- Proven and simple, but **limited adaptability and scaling for future hybrid complexity**



- Data-driven algorithms like Reinforcement Learning (RL) strategies enable handling of complex systems in efficient ways
 - Potential simplification of control structures
 - Potential real time adaptability
- Optimization of the interaction between ICE and drivetrain
- Currently a lot of research ongoing on integrating AI-Systems on control units
 - Technically feasible today – expected to become standard in future architectures
- Higher effort during development, but **essential for adaptability, efficiency, and future hybrid performance**

Overview/Schedule

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Task 6 – Task 8 on Model M1.1

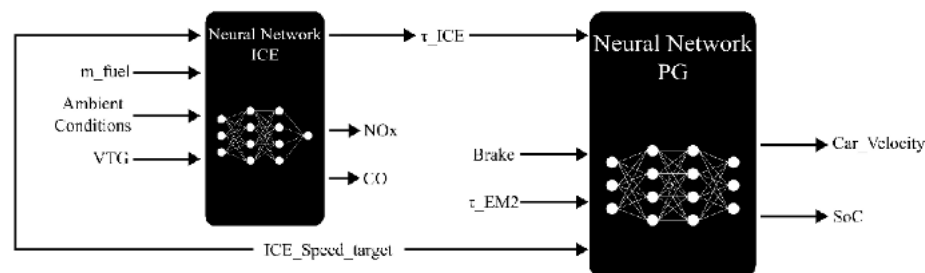
UPC



Description	Python Variable	Units	Range
Engine speed	ice_sp	rpm	0 – 4500
Fuel mass	mf	mg	3 – 70
Ambient temperature	T_amb_K	K	250 – 310
Ambient pressure	p_amb_bar	bar	0.8 – 1.1
Influence on turbine efficiency	vtg	%	23 – 90
ICE torque	torque	Nm	–50 – 300
Nitrogen oxides	Nox	mg/s	-
Nitrogen dioxide	NO2_out_m_gps	mg/s	-
Carbon monoxide	CO	mg/s	-
Carbon dioxide	CO2_out_m_gps	mg/s	-
EM2 torque	EM2	Nm	–421 – 421
Brake percentage	brk	%	0 – 100
State of Charge	soc	%	0 – 100
Car velocity	vel_out	km/h	-

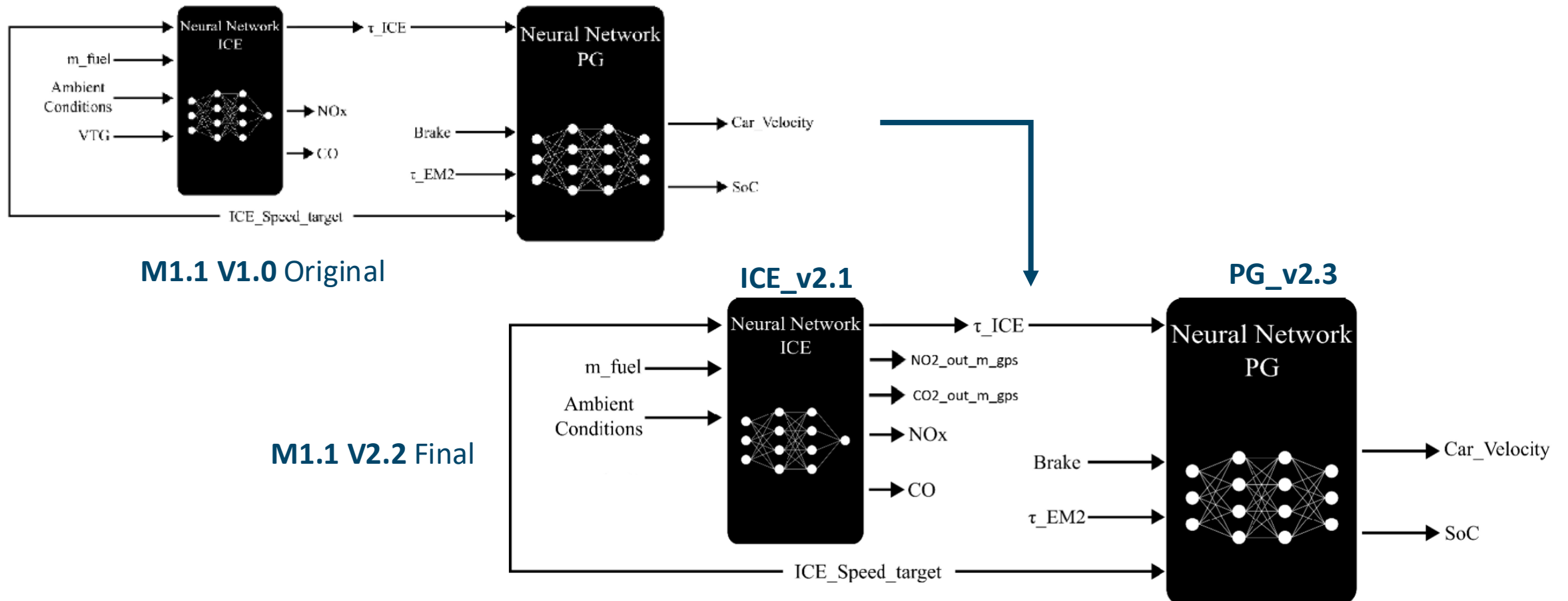
Work on Model M1.1

- **V1.0** Original
- **V2.0** (ICE Model):
If ice_sp < 900 --> mf, torque= minimum
- **V2.1** (ICE Model):
less inputs, more outputs – **ICE_v2.1**
- **V2.2** (PG Model):
New PG_v2.3



Task 6 – Task 8 on Model M1.1. MODEL ARCHITECTURE

UPC

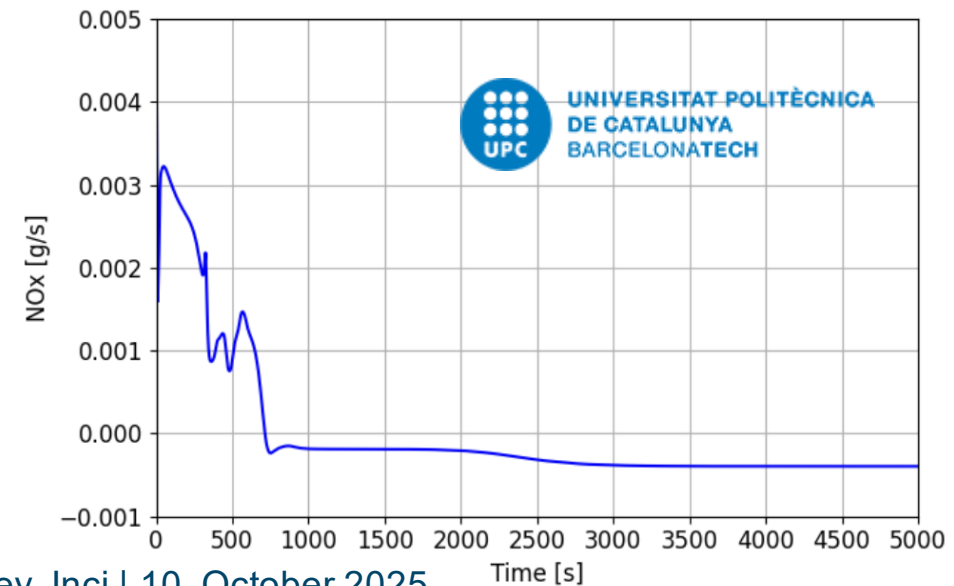
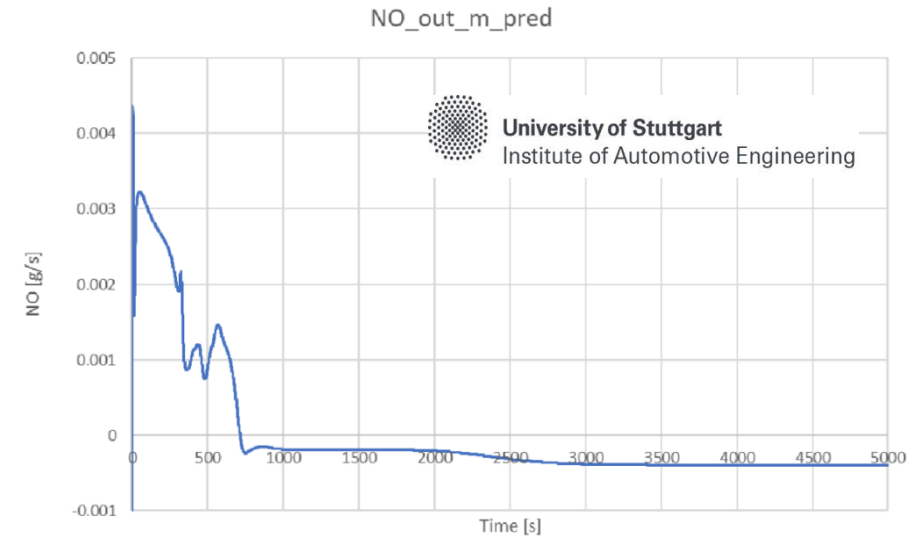


Task 6 – Task 8 on Model M1.1. DATA MODEL VALIDATION

UPC

Case 1. Low charge, low speed

- **ICE_Speed_rpm = 1000** (0 - 4500)
- **Fuelmass_mg = 3** (3 - 70)
- **EM2_Torque_Nm = 0** (-421 - 421)
- **Brake_frac = 0** (0 - 100)
- **SoC Start frac = 0.5** (0 - 1)

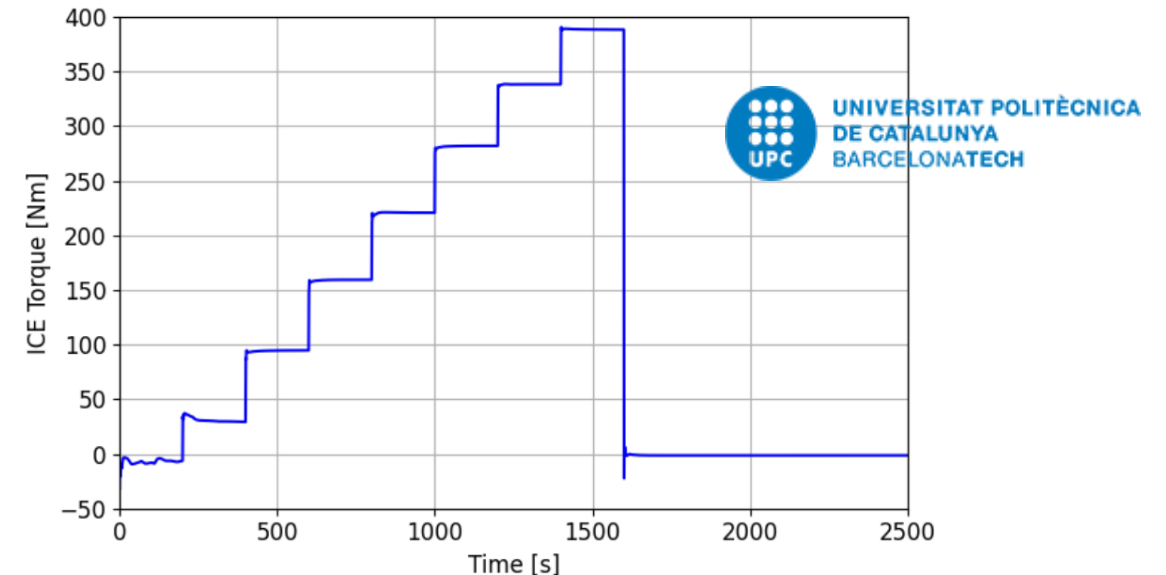
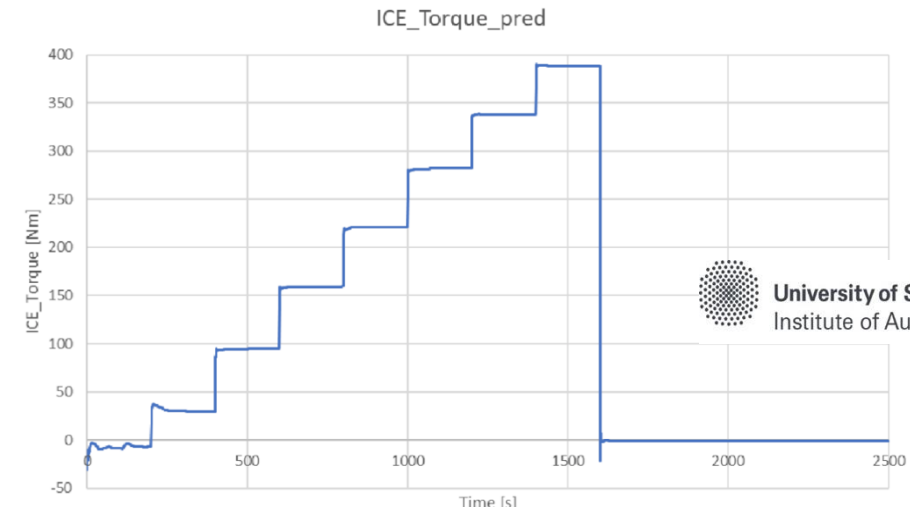


Task 6 – Task 8 on Model M1.1. DATA MODEL VALIDATION

UPC

Case 2. Low charge, average speed

- **ICE_Speed_rpm = 2500** (0 - 4500)
- **Fuelmass_mg = [3, 10, 20, 30, 40, 50, 60, 70, 5]** (3 - 70)
- **EM2_Torque_Nm = 0** (-421 - 421)
- **Brake_frac = 0** (0 - 100)
- **SoC Start frac = 0.5** (0 - 1)

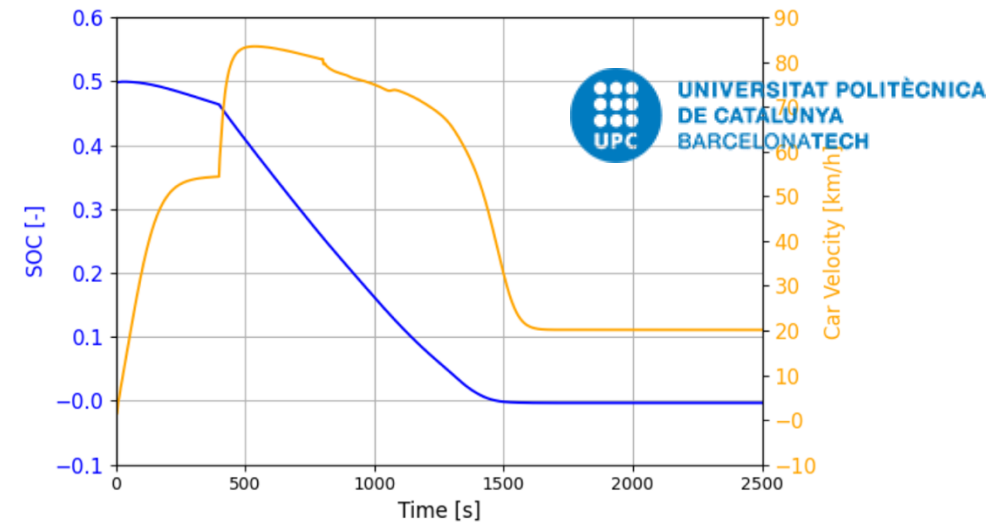
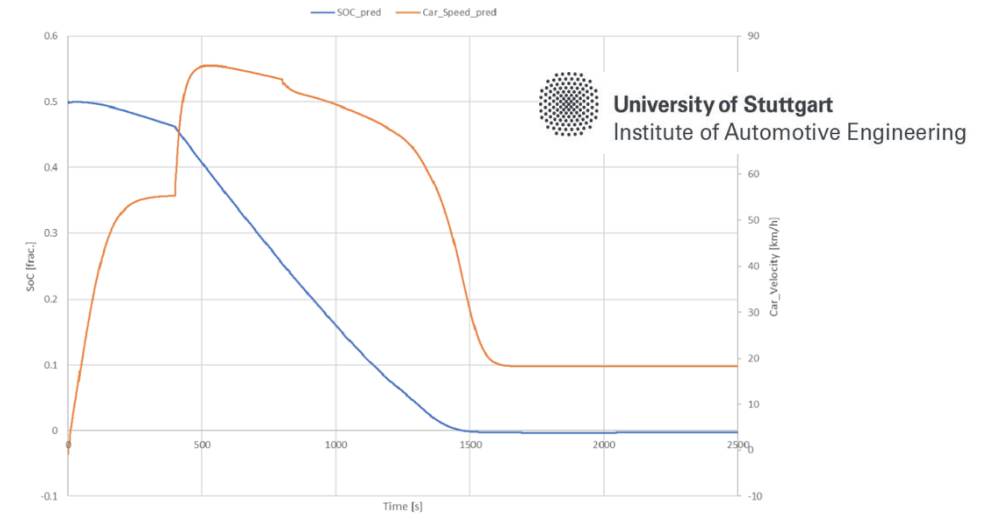


Task 6 – Task 8 on Model M1.1. DATA MODEL VALIDATION

UPC

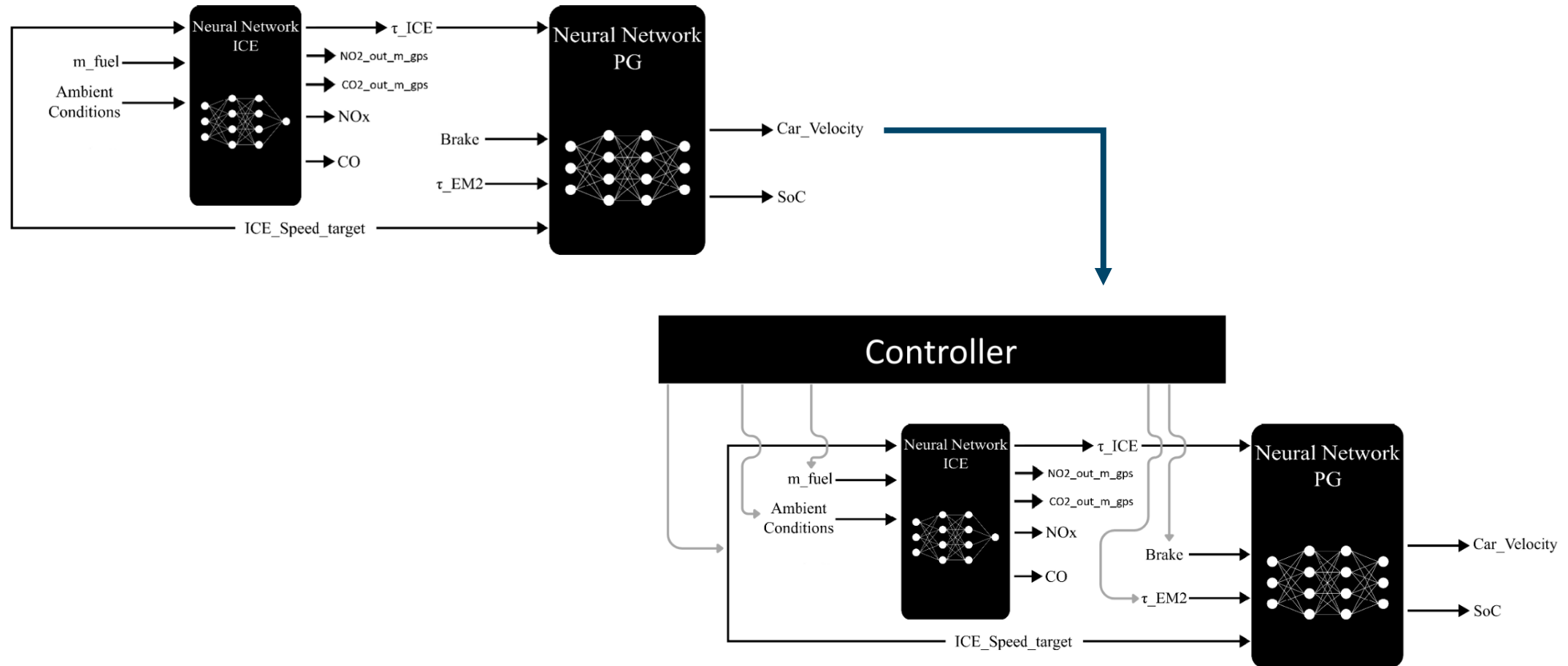
Case 3. Only electrical drive (ICE switch off)

- **ICE_Speed_rpm = 0** (0 - 4500)
- **Fuelmass_mg = 0** (3 - 70)
- **EM2_Torque_Nm = [50, 250, 400]** (-421 - 421)
- **Brake_frac = 0** (0 - 100)
- **SoC Start frac = 0.5** (0 - 1)



Task 6 – Task 8 on Model M1.1. CONTROL ARCHITECTURE

UPC



Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE



UPC

MBPC

TASK 6



University of Stuttgart
Institute of Automotive Engineering



Model 1.1

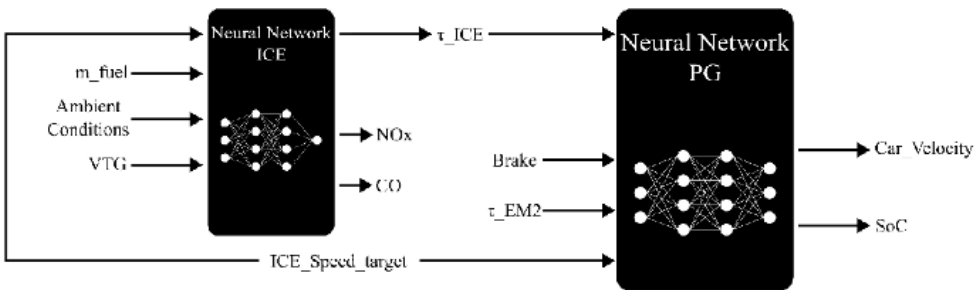
- MBPC allows:
 - Prototype a fast initial predictive controller
 - Reduce simulation costs for RL
 - Establish a reliable baseline before reinforcement learning (RL)

Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC

MBPC

TASK 6



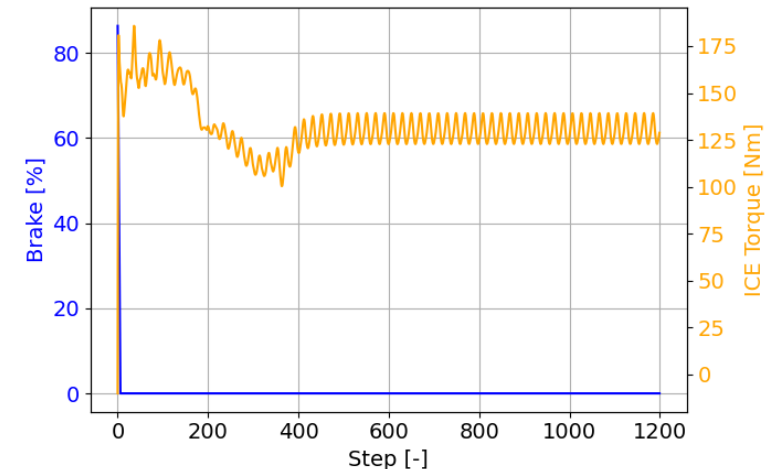
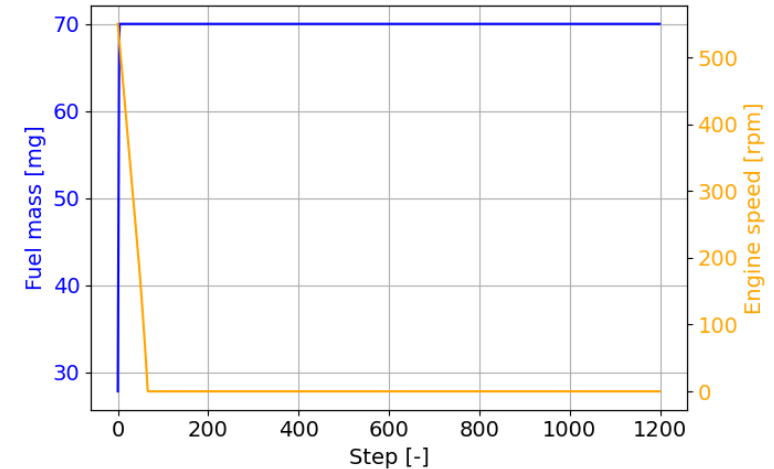
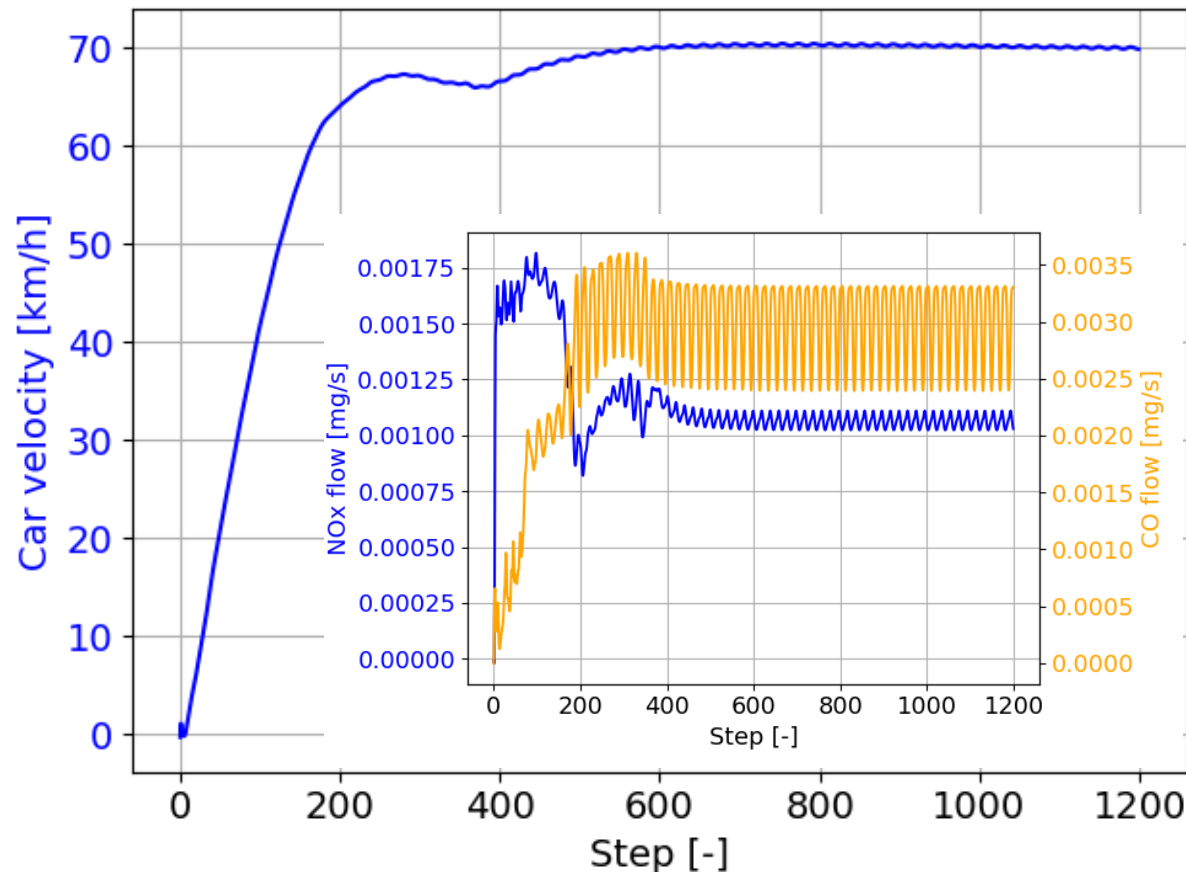
vel_out_ref = 70 $\rightarrow$ mf, vtg, brk, ice_sp

- Problem 1
 - Only fuel: Car Velocity reference 70km/h, EM2=0
 - **Model 1.1 v1.0** (VTG is still a control output)
 - Simulation 600 seconds @ 2Hz
 - Incremental controller
 - MSE loss $\text{step_loss} = \text{tf.square}(\text{vel_target} - \text{vel_out})$
 - Controller is a LSTM
 - TD learning
 - Actuators are bounded

Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC

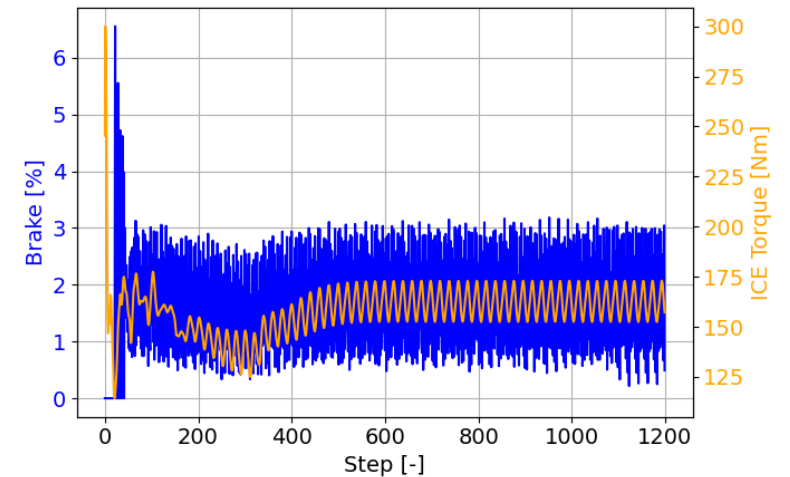
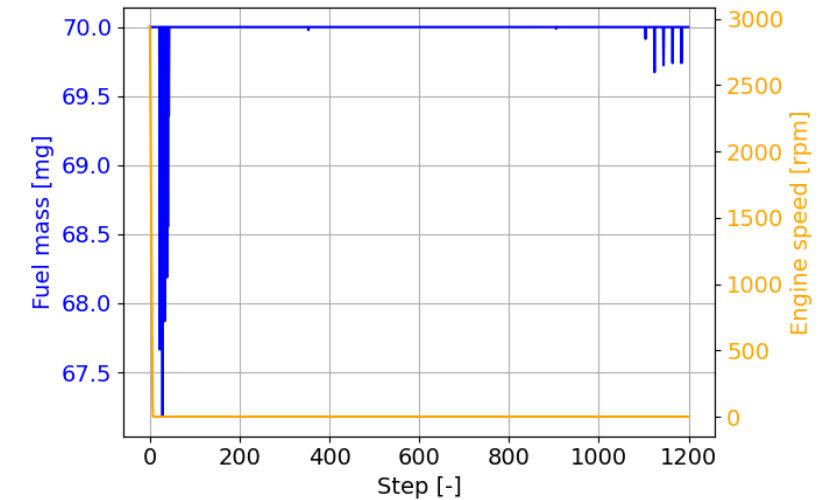
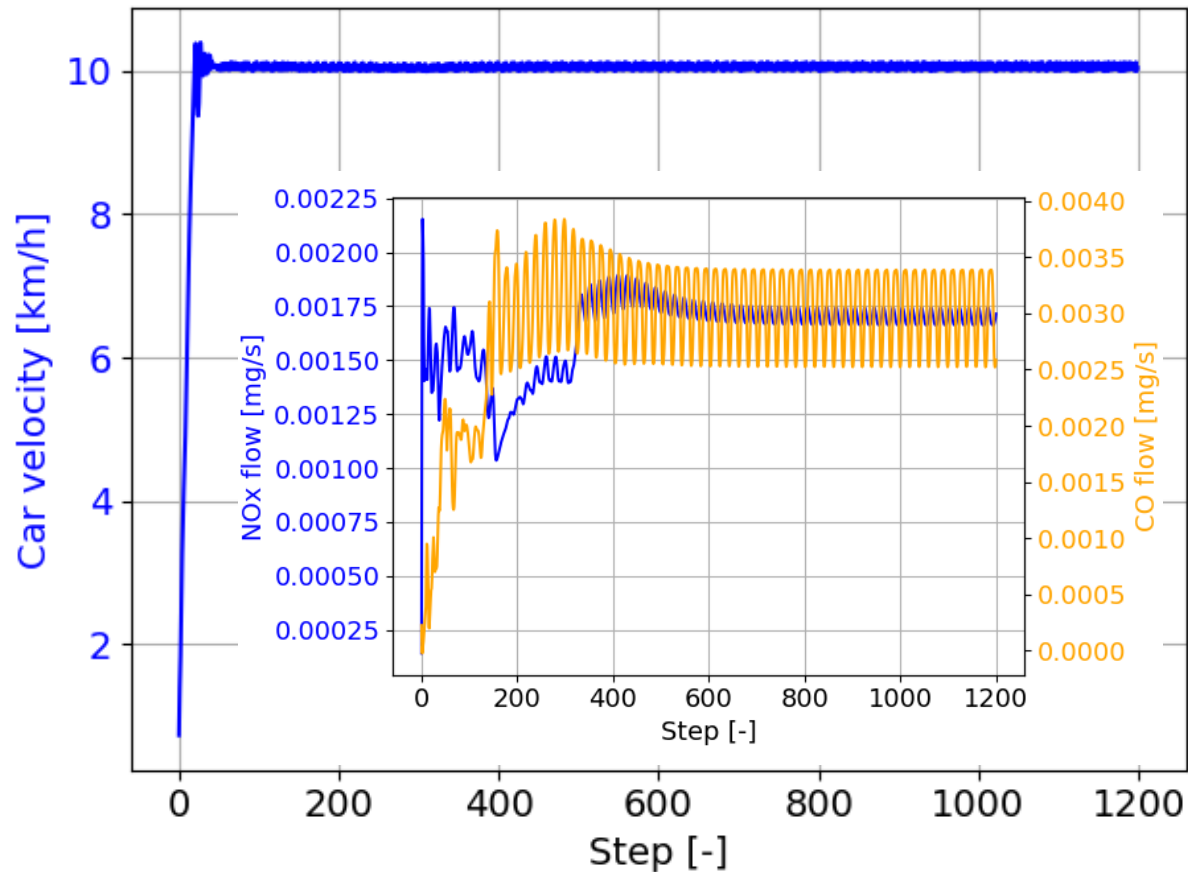
- Problem 1 . RESULTS
 - Only fuel: Car Velocity reference **70km/h**, EM2=0



Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC

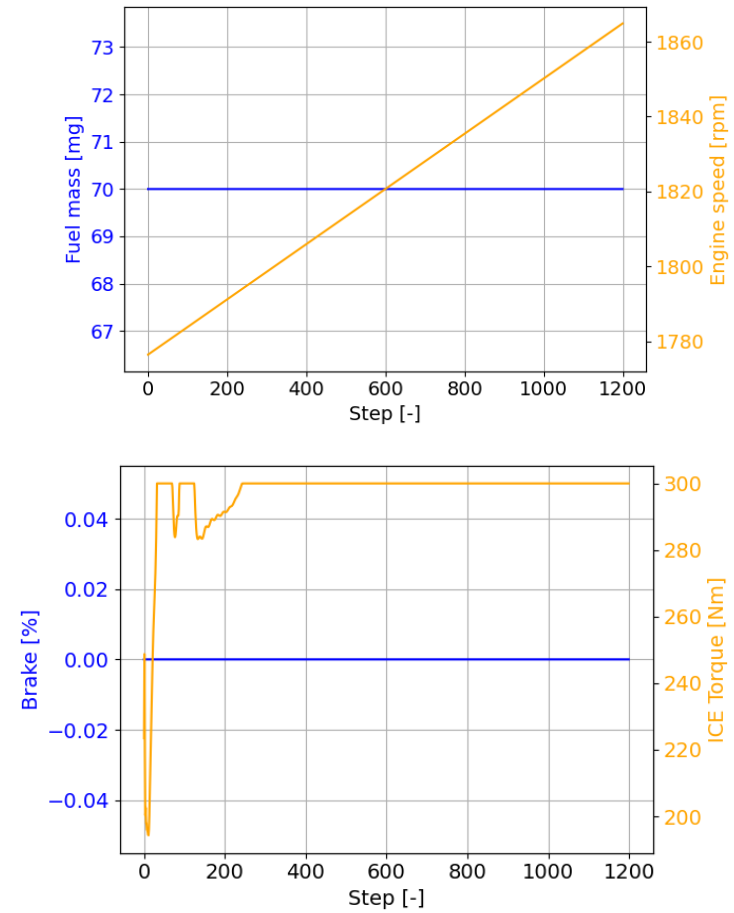
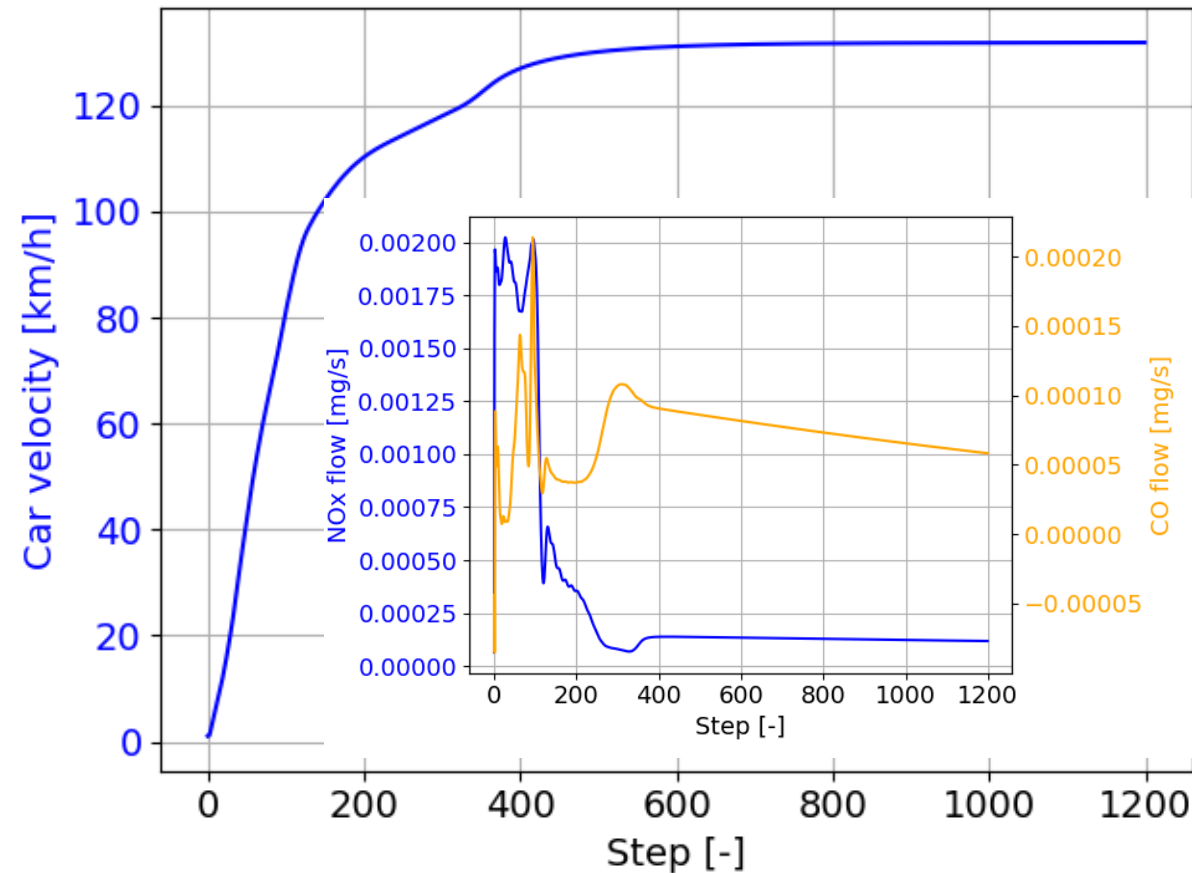
- Problem 1 . RESULTS
 - Only fuel: Car Velocity reference 10km/h, EM2=0



Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC

- Problem 1 . RESULTS
 - Only fuel: Car Velocity reference **130km/h**, EM2=0

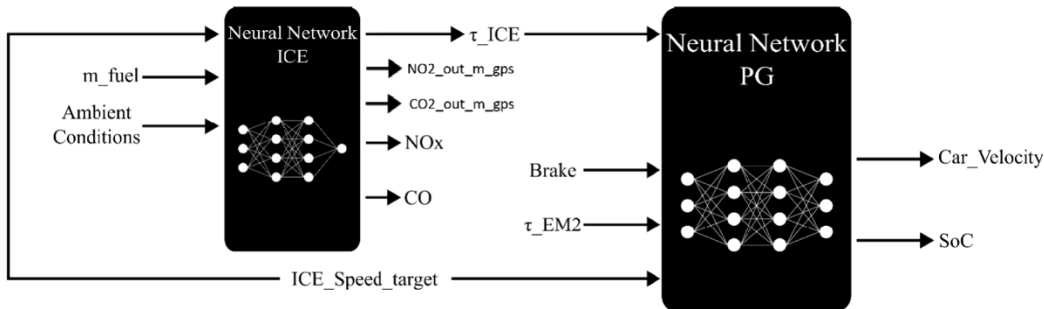


Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC

MBPC

TASK 6



$vel_out_ref = 70 \rightarrow mf, brk, ice_sp$

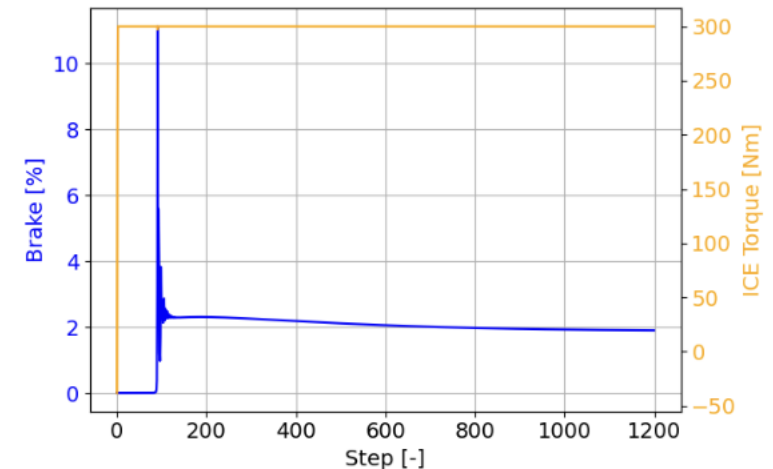
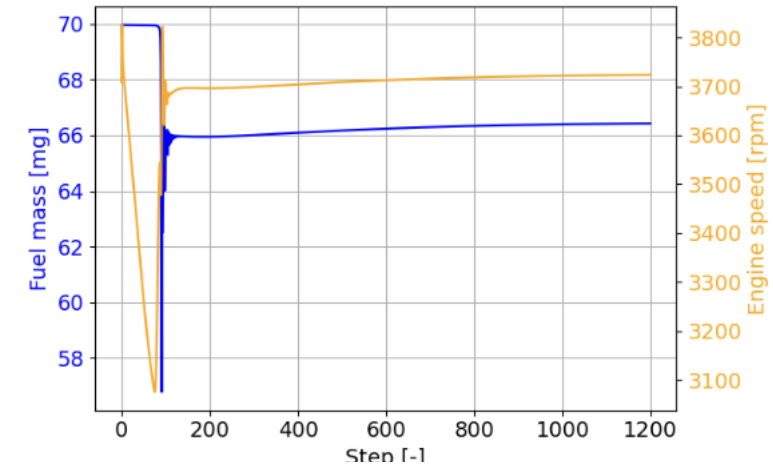
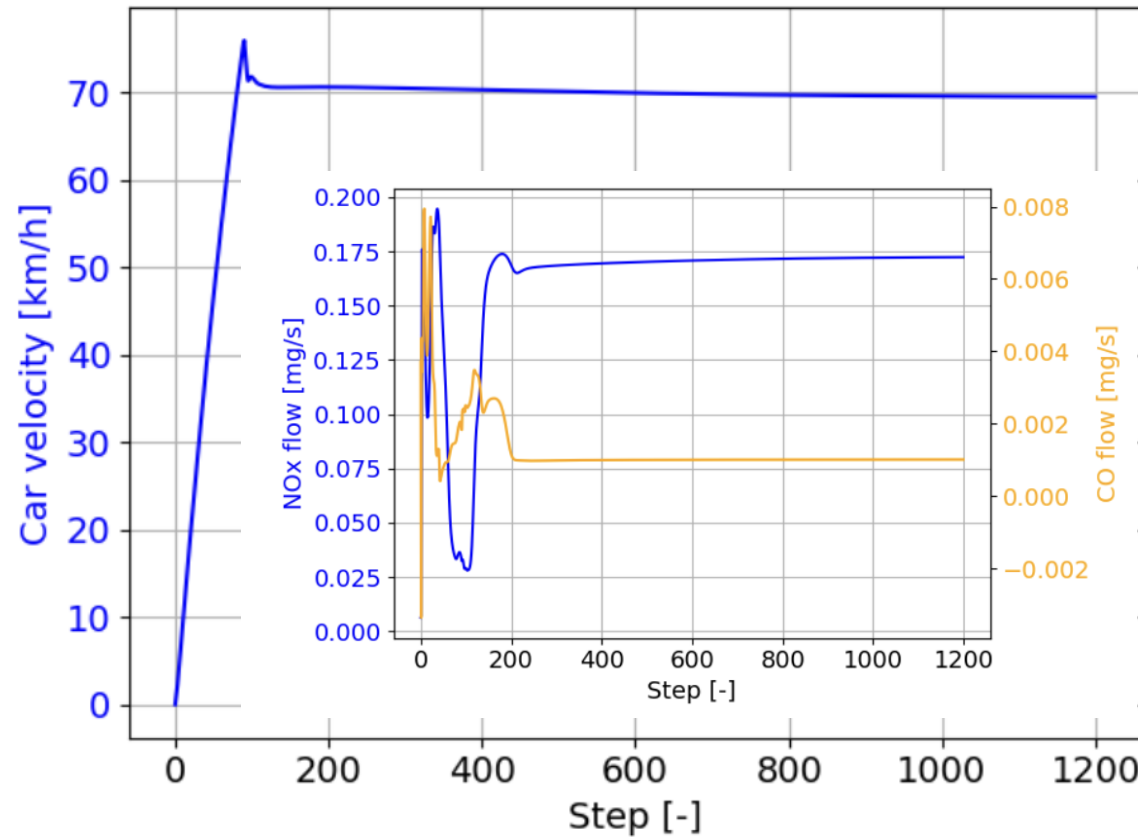
- Problem 2
 - Only fuel: Car Velocity reference 70km/h, EM2=0
 - **Model 1.1 v2.2**
 - Simulation 600 seconds @ 2Hz
 - Incremental controller
 - MSE loss $step_loss = tf.square(vel_target - vel_out)$
 - Controller is a LSTM
 - TD learning
 - Actuators are bounded

Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC

- Problem 2 . RESULTS

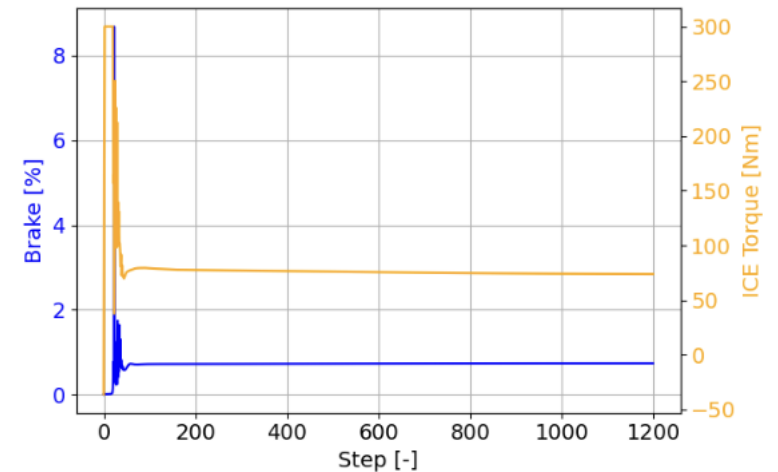
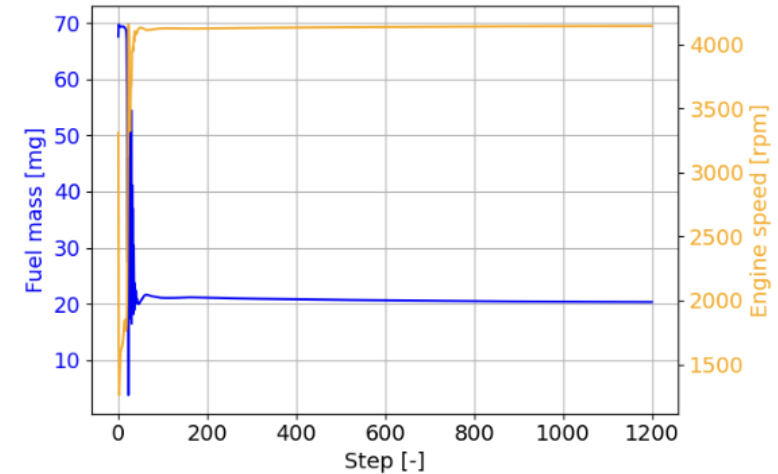
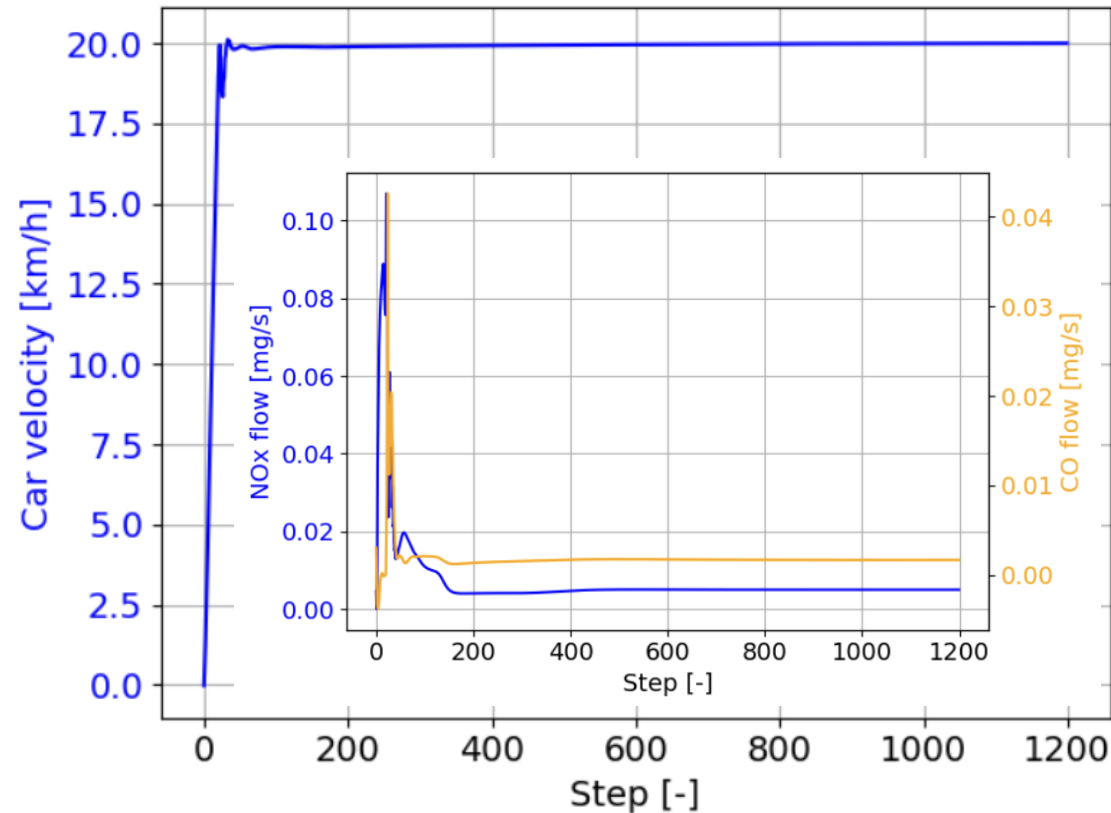
- Only fuel: Car Velocity reference **70km/h**, EM2=0



Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

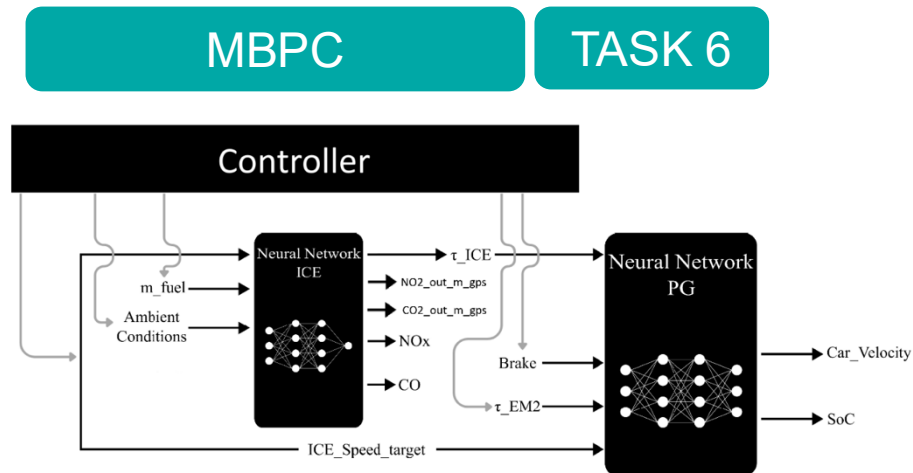
UPC

- Problem 2 . RESULTS
 - Only fuel: Car Velocity reference **20km/h**, EM2=0



Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC



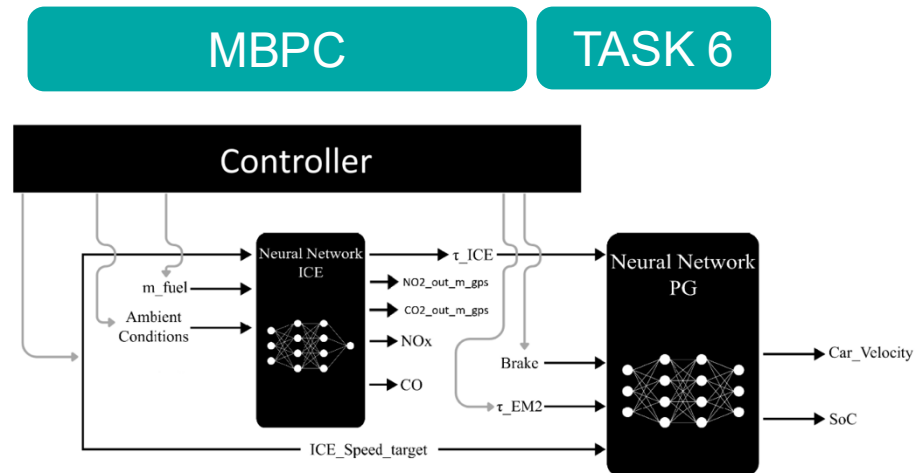
vel_out_ref = 70 → mf, brk, ice_sp

• Lessons learned

- Recurrent stateful (LSTM)
- Incremental controller
- Soft physical constraints
- Curriculum learning
- LR scheduling
- TD learning
- Realistic (no random) initialization

Task 6 – Task 8 on Model M1.1. LSTM-MBPC ARCHITECTURE

UPC



vel_out_ref = 70 → mf, brk, ice_sp, EM2

• NEXT STEPS

- **Hybrid model**
- Model 1.1 v2.2
- Simulation 600 seconds @ 2Hz
- Incremental controller
- **MSE loss**
- Controller is a LSTM
- TD learning
- Actuators are bounded

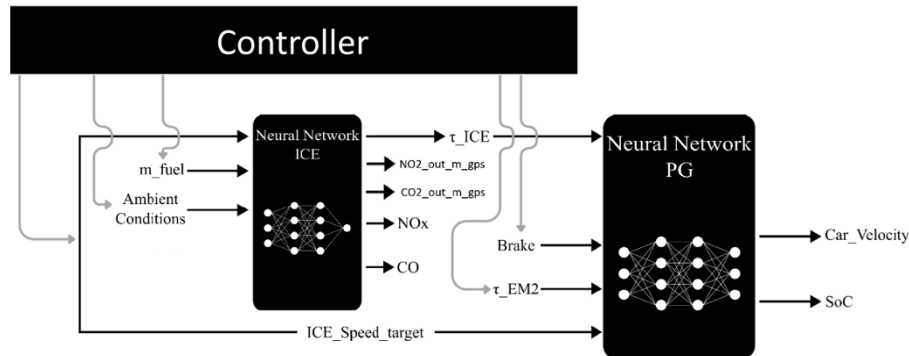
$$\min_{u_0, \dots, u_{N_p-1}} J = \sum_{k=0}^{N_p-1} \left[w_v (v_{\text{ref}}(k) - v(k))^2 + \sum_{g \in \{NO_x, NO_2, CO, CO_2\}} w_g \dot{m}_g(k) \right]$$

Task 6 – Task 8 on Model M1.1. RL ARCHITECTURE

UPC

Complete pipeline

TASK 7



vel_out_ref = 70 → mf, brk, ice_sp, EM2

RL strategy

TASK 8

• CURRENT STATE

- Recurrent TD3 RL because POMDP problem
- Recurrent Actor (LSTM)
- Recurrent Critic (LSTM)
- Replay buffer on time-windows (long trajectory)
- Burn-in/unroll (init weights) and skip-tail (early ending)
- Off-policy training
- Curriculum learning to be validated
- Actor-learner in parallel (time saving)

Agenda



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Next Steps

GOALS UNTIL THE NEXT PA

FZA

- ...

IFS

- Finalizing Calibration of EA 288evo simulation model
- Generating data with the EA 288evo model to train the final Neural Network of the ICE and Powertrain
- Setting up and tune a rule-based operating strategy in the simulation environment
- Starting documentation for the final report

UPC

- Extending LSTM-MBPC control architecture to new situations
- Fitting RL Recurrent TD3 control architecture

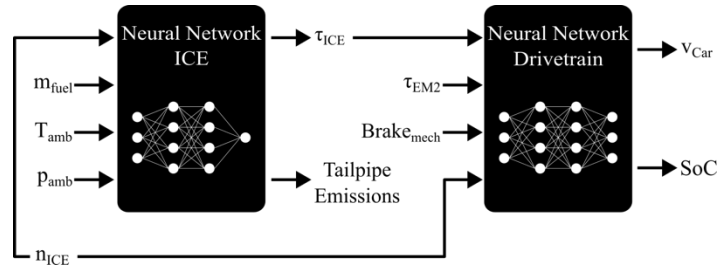
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NEXT PA MEETING



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POSSIBLE FUTURE RESEARCH ASPECTS - PHYSICAL SIMULATION AND AI PERSPECTIVE



Focus in this Project

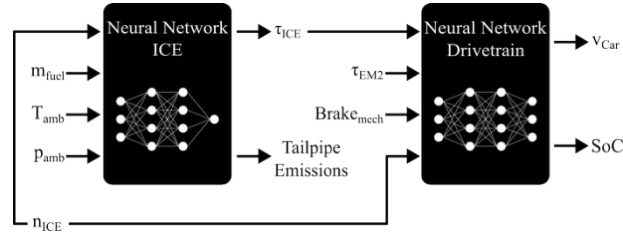
- Operating strategy of hybrid powertrain
- Power distribution between powertrain components
- **Proof of AI - concept on system level operating strategy** using the example of a powersplit hybrid system
- System complexity must not be underestimated

Possible future Research

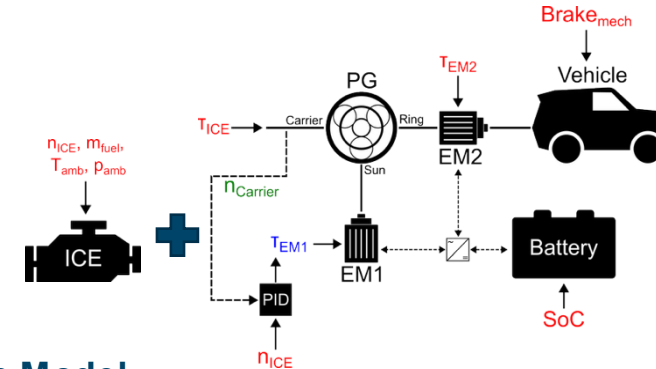
- Deepening the complexity by increasing optimization parameters
- Currently 4 parameters that have to be optimized by the RL agent
- Future approach could include more ICE relevant parameters like EGR, VTG, SOI, ...
- **What does increasing the number of optimization parameters mean for the physical and AI model?**

Challenges and Limits of Methodology in future Research

WHAT DOES INCREASING THE NUMBER OF OPTIMIZATION PARAMETERS MEAN FOR THE PHYSICAL AND AI MODEL?



Training



AI – Model

- More data necessary to cover the increasing number of possible parameter combinations (disproportionately)
- Example:
 - 4 parameters $\rightarrow$ >750k operating points
 - 5 parameters $\rightarrow$ >1mio. operating points
- Exact number is dependent on system complexity and application

Physical Simulation Model

- More necessary data results in higher computational effort and license costs
 - Increasing number of optimization parameters leads to:
 - Strongly coupled effects within one system (ICE)
 - Risk of interactions outside valid range of numerical models (instability, unrealistic states)
 - Frequent numerical breakdowns due to non-combusting or non-feasible operating points
 - Exponentially increasing simulation time for calibration and validation
- Increasing optimization parameters drive models towards instability and non-feasible operating points – **enormous countermeasures are needed** (pre-screening, fallback or simple models, extensive measurement support)

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